

Scalable ML Assignment

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1 Log Mining and Analysis

Table 1: Total requests and unique hostnames by UK sector.

Sector	Total Requests	Unique Hosts
Academic	25009	1022
Company	22225	1116
Government	195	15

For the `.ac.uk` hosts, the institution domain is extracted by first splitting each hostname using the full stop delimiter. For example, the Sheffield hostname, `www.shef.ac.uk`, is split into `["www", "shef", "ac", "uk"]`. The final three labels are concatenated together, resulting in `shef.ac.uk`, which represents the institution domain. Different hosts from the same institution (`cs.shef.ac.uk`, `www.shef.ac.uk`, etc.) are grouped together within the Sheffield domain.

Listing 1 shows the Spark implementation of the method described above.

Listing 1: Institution domain extraction and aggregation.

```
academic_hosts = logs.filter(col("host").endsWith(".ac.uk"))

split_host = split(col("host"), "\\.")

academic_domains = academic_hosts.select(
    col("host"),
    concat_ws(".",
        element_at(split_host, -3),
        element_at(split_host, -2),
        element_at(split_host, -1)
    ).alias("institution")
)

institution_counts = academic_domains.groupBy("institution") \
    .count() \
    .withColumnRenamed("count", "total_requests")
```

The total number of requests from the University of Sheffield domain is 623. The seven institutions with more requests than Sheffield, and their respective total requests, are shown below:

Table 2: UK academic institutions with more requests than the University of Sheffield.

Institution	Total Requests
<code>hensa.ac.uk</code>	4257
<code>rl.ac.uk</code>	1158
<code>ucl.ac.uk</code>	1036
<code>man.ac.uk</code>	921
<code>ic.ac.uk</code>	851
<code>soton.ac.uk</code>	808
<code>bham.ac.uk</code>	629

The nine most active UK (`.co.uk`) domains and their respective total requests are displayed in Figure 1. A bar chart is chosen to compare request counts across discrete company categories. It clearly shows the top nine companies alongside all remaining companies combined, with contrasting colours used to highlight the comparison and improve readability.

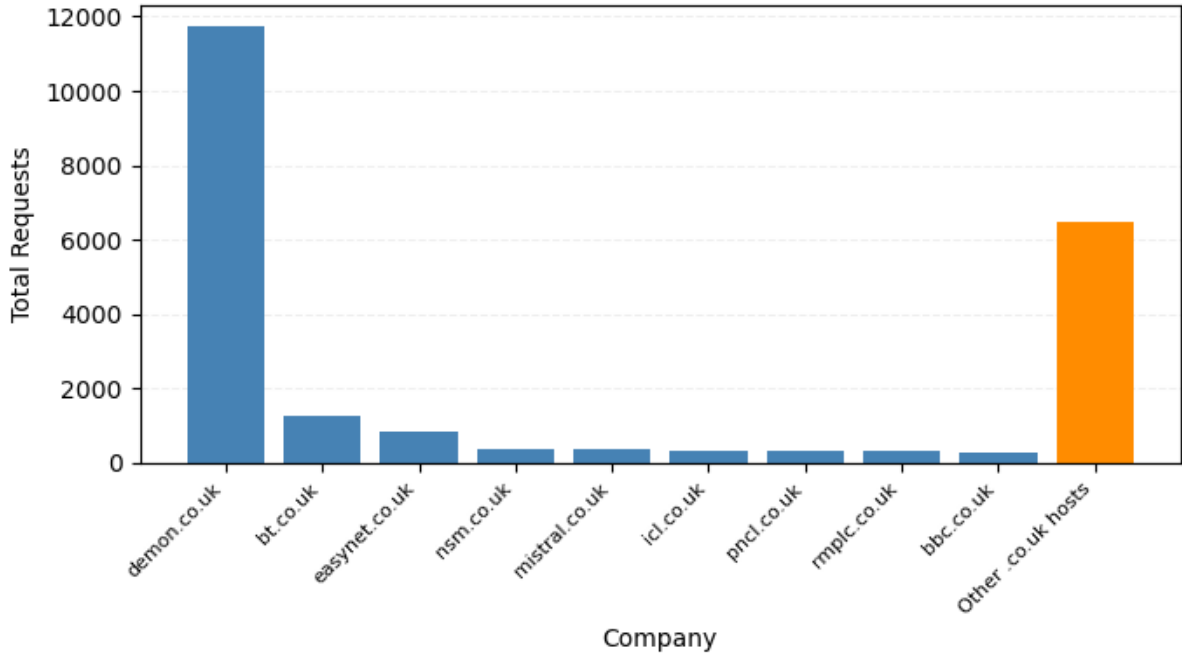


Figure 1: Total requests from the nine most active UK company domains, compared with the combined total for all remaining UK companies.

The most active academic institution is `hensa.ac.uk`, while the most active company domain is `demon.co.uk`. Figure 2 shows the hourly request patterns for these domains and all `.gov.uk` hosts combined. The heatmaps show that access activity is typically concentrated during daytime and evening hours. The `demon.co.uk` has the densest activity overall, whereas the combined `.gov.uk` requests are much more infrequent.

2 Scalable Logistic Regression and Generalised Linear Models

The traffic vehicle-count dataset is split by year into training, validation, and test sets. The sample sizes and their corresponding years are shown in Table 3.

Table 3: Sample sizes of training, validation and test sets

Set	Years	Sample Size
Training	2000–2021	4655073
Validation	2022–2023	306804
Test	2024	151848

The `regParam` hyperparameter is tuned over five values, each with five random subsets that sample 80% of the training set. The random seeds used are [17677, 17678, 17679, 17680, 17681] (from student number 250117677).

The resulting mean validation MSE of each `regParam` value is shown in Fig. 3. The optimal value is `regParam` = 0.001, which has the lowest mean validation MSE in Fig. 3 ($813\,706.18 \pm 237.58$). Increasing `regParam` from 0.001 to 0.01 and 0.1 causes a sharp increase in MSE, which plateaus for larger values. This suggests that stronger regularisation causes the Poisson regression model to underfit.

Using the optimal value of `regParam` = 0.001, the model is retrained on the combined training and validation sets. The final model is tested on the test set, achieving a mean validation MSE of 787 575.11.

The learned coefficient vector is reported in Appendix A.

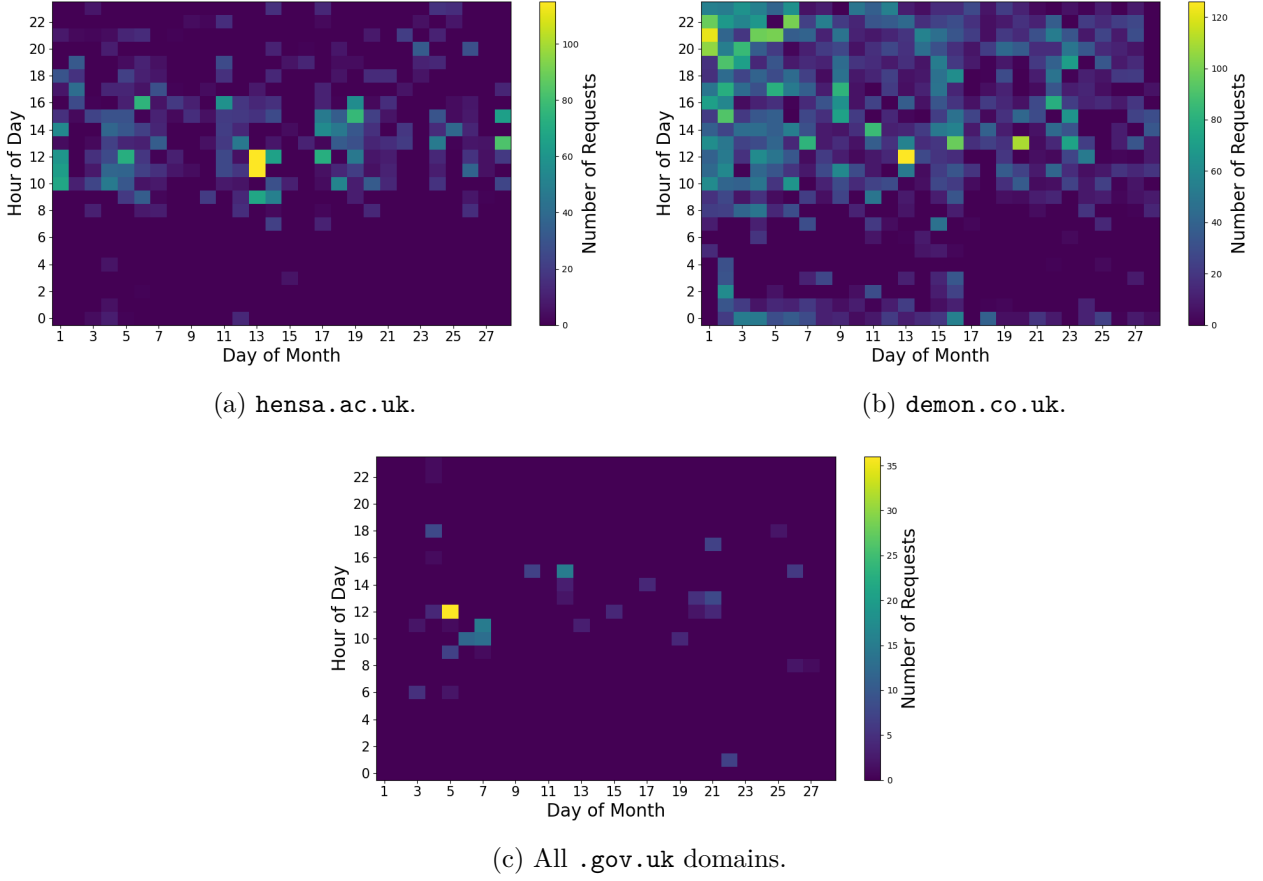


Figure 2: Hourly access patterns across days 1–28. Colour intensity represents the number of requests.

A logistic regression model is trained using the same features. The same values of `regParam` are tested, as well as `elasticNetParam` values of $[0.0, 0.2, 0.5, 0.8, 1.0]$. Five random training subsets are created to determine the mean validation accuracy for each combination of `regParam` and `elasticNetParam`, which is shown in Fig. 4.

3 Scalable Decision Trees, Random Forests, and Gradient Boosting

Table 4: Optimal hyperparameters selected by 3-fold cross-validation on the 2% sample.

Model	Parameter 1	Parameter 2	Parameter 3
Random Forest	<code>maxBins</code> = 64	<code>maxDepth</code> = 7	<code>numTrees</code> = 20
Gradient-Boosted Tree	<code>maxBins</code> = 32	<code>maxDepth</code> = 7	<code>maxIter</code> = 40

Table 5: Training and test accuracy of the tuned models on the sampled HIGGS dataset.

Model	Training Accuracy	Test Accuracy
Random Forest	0.701	0.692
Gradient-Boosted Tree	0.747	0.720

The full HIGGS dataset is split into a training set of 7 698 149 and a test set of 3 301 851 samples, which are used to evaluate both the Random Forest and Gradient-Boosted Tree models. The resulting performance of the models is shown in Table 6.

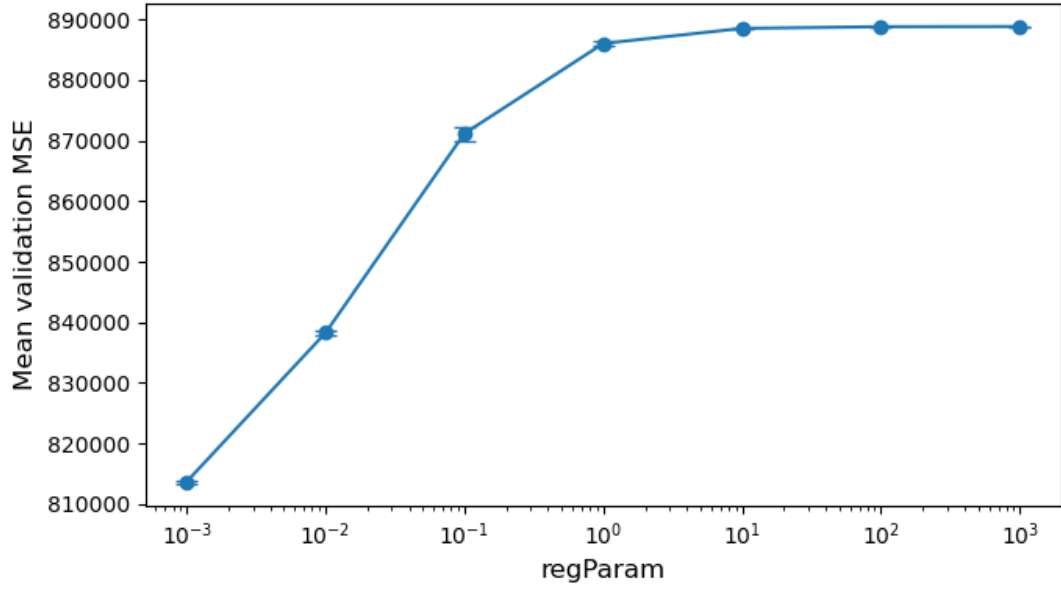


Figure 3: Mean validation MSE as a function of `regParam`. Error bars show the standard deviation, calculated over five random subsets per `regParam` value.

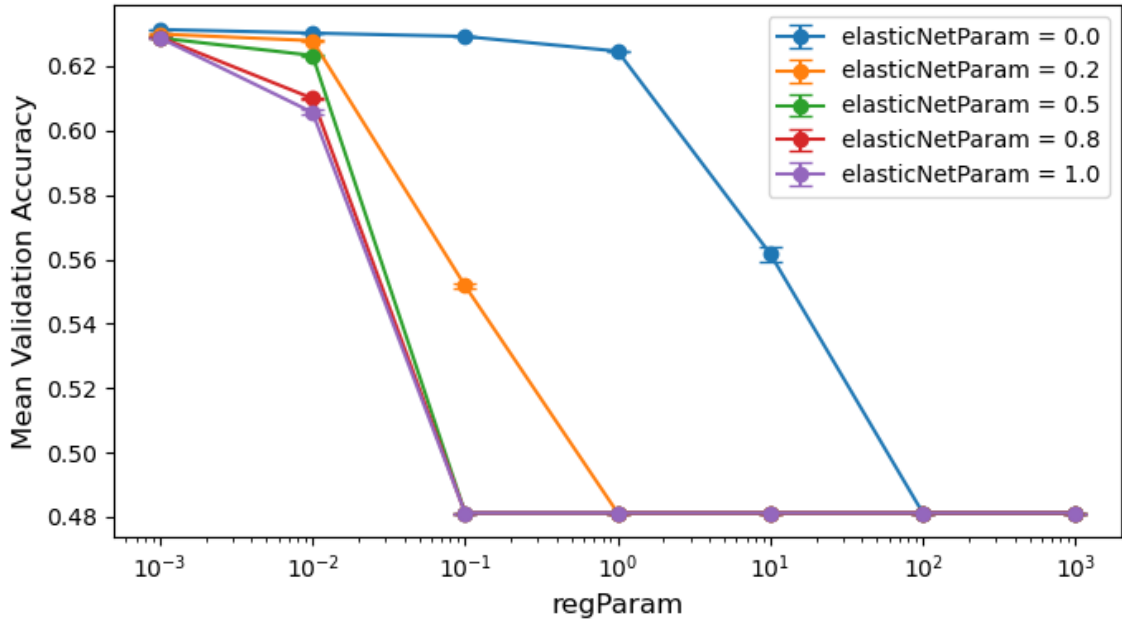


Figure 4: Mean validation accuracy as a function of `regParam`, combined with different values of `elasticNetParam`. Standard deviation values are calculated by evaluating each combination of `regParam` and `elasticNetParam` over five random subsets of the training data.

Table 6: Training and test accuracy of the tuned models on the full HIGGS dataset.

Model	Training Accuracy	Test Accuracy
Random Forest	0.692	0.691
Gradient-Boosted Tree	0.726	0.726

Table 7: Recommendation performance for the two ALS settings across chronological train-test splits.

Training Split	ALS Setting 1			ALS Setting 2		
	RMSE	MSE	MAE	RMSE	MSE	MAE
40%	0.803	0.645	0.616	0.822	0.675	0.635
60%	0.777	0.603	0.591	0.792	0.628	0.605
80%	0.797	0.635	0.604	0.810	0.656	0.617

4 Movie Recommendations and Cluster Analysis

Appendix

Learned model coefficients.

Table 8: Top five user-cluster sizes for each chronological training split using ALS Setting 2 user factors and k-means with $k = 25$.

Training Split	Cluster Size				
	Rank 1	Rank 2	Rank 3	Rank 4	Rank 5
40%	4543	4373	4353	4291	4052
60%	5975	5968	5941	5881	5565
80%	8455	7990	7866	7002	6699

Table 9: Top ten most popular genres among highly rated movies in the largest user cluster for each chronological training split.

Rank	40% Split		60% Split		80% Split	
	Genre	Count	Genre	Count	Genre	Count
1	Drama	449	Drama	167	Drama	591
2	Comedy	211	Comedy	62	Comedy	293
3	Romance	125	Romance	34	Romance	178
4	Thriller	112	Documentary	34	Thriller	138
5	Crime	101	Thriller	31	Action	128
6	Action	87	Crime	30	Crime	123
7	Adventure	80	War	29	Adventure	110
8	War	75	Adventure	28	Documentary	102
9	Documentary	52	Action	25	War	96
10	Mystery	50	Mystery	19	Mystery	59